

Drake Rundell

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Professional Summary

Full-stack software engineer with five-plus years delivering production systems across web applications, cloud infrastructure, data pipelines, and embedded hardware. Built a React/TypeScript and Flask application from the ground up at General Motors now used by approximately 200 engineers, leading and owning the complete delivery cycle from requirements through development, deployment and operations.

Skills

- **Technical Skills:** Python, TypeScript, JavaScript, C++, C, Java, React, Node.js, Flask, LVGL, MUI, Vite, MicroPython, PostgreSQL, Neo4j, SQLite, Databricks, Delta Lake, Docker, Kubernetes, Azure
- **Tools & Platforms:** Git, MQTT, PlatformIO, ESP-IDF, ESP32, KiCAD, Fusion 360, Raspberry Pi, Azure CLI, MATLAB
- **Architecture & Design:** Graph database modeling (Neo4j), data pipeline design, IoT system architecture, PCB layout, embedded firmware state machines
- **Collaboration:** Stakeholder alignment, executive communication, technical documentation, vendor management

Experience

General Motors

February 2023 – Present

Full-Stack Software/System Engineer, SDV Embedded Platforms

Warren, MI | Remote

- Built and lead a React/TypeScript and Flask (Python) web application from the ground up, now used by approximately 200 engineers to visualize and navigate Software-Defined Vehicle (SDV) software and system architecture. Own the complete delivery cycle — requirements, design, implementation, deployment, and operations — on a 2-person team.
- Led technical evaluation of the graph visualization library stack, assessing open-source and commercial options across clustering performance, render scale, bundle size, and API ergonomics at tens of millions of nodes. Drove the final adoption decision and delivered results to production.
- Architect Neo4j graph schema and relationship ontology for SDV system and software dependencies, scaling to tens of millions of nodes and edges representing the full SDV architecture. Serve as the internal graph database architect and maintain a direct vendor support relationship with Neo4j.
- Designed and implemented a multi-source Python ingestion pipeline pulling from planning systems, test result databases, and systems engineering model sources. Normalized heterogeneous schemas and unified data into PostgreSQL and Neo4j via Databricks on Azure.
- Operate and maintain automated nightly Databricks pipelines on Azure serving dual customer segments: application users consuming data through the frontend, and raw Data Mesh consumers accessing data directly. Responsible for cloud infrastructure deployment, reliability, and uptime of the full application stack.

General Motors

July 2021 – February 2023

Design Release Engineer, Air Delivery Systems

Warren, MI

- Authored two granted U.S. patents (**US Patent No. 11,933,256** and **US Patent No. 11,852,109**) concerning novel approaches to vehicle gas distribution in intake manifold runners, filed independently of any expectation of the role.
- Owned design release for intake manifolds and engine covers across multiple production and development programs. Managed relationships with 3 direct suppliers and drove cross-functional design reviews with engine architecture, vehicle program, and validation teams.

- Developed tooling, fixtures, and additive manufacturing solutions for precision electromechanical assemblies; led development of a plastic additive area enabling low-cost rapid tooling production with a path toward production hardware.

Projects

Goal Siren — IoT NHL Goal Alert System

Python, Flask, React, TypeScript, MUI, C++, ESP32, Docker, MQTT, SQLite, KiCAD, Pillow, Raspberry Pi

- Designed and built a complete end-to-end IoT system from hardware to UI: custom 2-layer KiCAD PCB, ESP32 C++ firmware with interrupt-driven MQTT subscribe/publish state machine, REST API backend with 40+ endpoints, React and TypeScript admin frontend with MUI v6, and Docker-containerized deployment on a Raspberry Pi Zero.
- Implemented a real-time event detection pipeline integrating with the NHL live data API through a Mosquitto MQTT broker, achieving under 3-second end-to-end latency from API event to physical hardware response.
- Built a dynamic PNG card generation system (240x320 px) using Pillow for on-device display content; SQLite backend for full CRUD across teams, players, and game state; TypeScript interfaces mirroring API response shapes for type-safe consumption across all endpoints.

RPI Matrix — NHL LED Scoreboard

TypeScript, React, Node.js, Vite, C++, MicroPython, ESP32, MQTT, SQLite, Server-Sent Events

- Built a full-stack TypeScript system: Node.js backend, React and Vite frontend, MicroPython firmware on ESP32 communicating over MQTT, and native C++ canvas bindings for sub-100ms RGB frame generation streamed directly to dual HUB75 LED panels.
- Implemented Server-Sent Events for real-time browser preview of the LED display without polling. Designed an async display state machine with AbortController for clean concurrent request handling and interrupt-driven display cycle management.
- Built a data-heavy dashboard featuring team stats, game scores, and skater and goalie data stored across SQLite. Features include drag-and-drop display ordering, team selection checklists, a settings panel, and data import controls.

PlantNode — Environmental Monitor

C++, Python, ESP32-C3, LVGL, KiCAD, Fusion 360, I2C, UART, ADC, PlatformIO

- Designed a custom 2-layer KiCAD PCB and enclosure for an ESP32-C3 sensor node with a multi-sensor array (ADC, time-of-flight, light, soil moisture) over I2C, and an 8-byte UART telemetry protocol with XOR checksum.
- Built a parametric plant-rendering engine in C++ using LVGL v9 — all visuals procedurally generated from rect primitives with per-leaf sway animation within a 16ms frame budget; Python tooling mirrors firmware geometry for desktop preview and automated stage-table sync.

Education

Georgia Institute of Technology

Master of Science, Computer Science (Computing Systems) — OMSCS

Expected May 2027

University of Michigan — Ann Arbor

Bachelor of Science in Engineering, Aerospace Engineering

May 2020